

FERGANA, Uzbekistan

WMO: 386180

Lat: 40.367N

Lon: 71.750E

Elev: 604

StdP: 94.28

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.7	-6.1	-11.7	1.5	-7.2	-9.2	1.8	-4.3	5.8	4.1	4.3	2.2	1.1	120	0.396

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.5	37.1	20.7	35.9	20.5	34.7	20.2	22.9	33.5	22.1	32.8	21.4	31.9	2.1	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.6	15.4	28.2	18.6	14.5	27.6	17.7	13.6	27.2	71.2	33.8	68.1	33.0	65.4	32.0	26.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.2	5.1	4.0	-10.3	39.7	3.9	1.1	-13.0	40.5	-15.3	41.2	-17.4	41.8	-20.2	42.6		
			-10.7	24.9	3.8	0.9	-13.5	25.5	-15.7	26.0	-17.8	26.5	-20.6	27.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	15.1	0.9	3.3	10.3	16.5	21.6	26.0	28.0	26.7	21.9	15.1	7.9	2.2		
	DBStd	10.11	3.52	4.18	4.21	3.95	3.13	2.57	2.14	2.37	2.97	3.76	3.96	3.18		
	HDD10.0	854	283	189	48	4	0	0	0	0	0	6	84	241		
	HDD18.3	2228	541	421	248	79	9	0	0	0	7	112	312	499		
	CDD10.0	2713	0	2	58	198	359	479	558	517	356	163	22	0		
	CDD18.3	1043	0	0	1	23	109	229	299	258	113	11	0	0		
	CDH23.3	12237	0	0	6	210	1085	2752	3879	3044	1149	111	1	0		
	CDH26.7	6137	0	0	0	48	400	1417	2168	1635	451	18	0	0		
(14) Wind	WSAvg	1.5	1.1	1.3	1.6	1.8	1.9	1.9	1.8	1.7	1.6	1.4	1.3	1.2		
(15) Precipitation	PrecAvg	175	19	23	26	19	20	9	4	2	4	17	16	16		
	PrecMax	343	72	93	68	70	87	33	26	10	24	135	83	50		
	PrecMin	82	0	0	1	0	2	0	0	0	0	0	0	0		
	PrecStd	61	14	20	16	14	16	9	5	3	6	26	17	15		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.5	16.6	24.1	30.4	34.2	38.1	39.2	37.9	34.1	29.1	21.8	12.4		
		MCWB	6.7	9.1	13.4	17.7	19.1	19.9	21.0	21.5	20.1	17.5	14.0	7.4		
	2%	DB	9.1	13.1	21.5	28.0	32.4	36.2	37.8	36.2	32.5	26.2	18.6	10.1		
		MCWB	5.4	7.8	12.5	16.7	18.6	19.9	20.7	21.0	19.1	16.1	11.9	6.2		
	5%	DB	7.2	11.0	19.5	25.8	30.8	34.8	36.2	35.1	31.1	24.1	16.0	8.5		
		MCWB	4.4	6.8	11.9	15.8	17.9	19.7	20.5	20.7	18.4	14.8	10.7	5.2		
	10%	DB	5.8	9.2	17.4	23.7	28.9	33.2	35.0	33.9	29.4	22.1	13.9	7.0		
		MCWB	3.6	5.8	11.0	15.0	17.5	19.2	20.3	20.3	17.5	13.9	9.4	4.4		
	0.4%	WB	7.2	9.8	14.6	19.4	21.6	23.2	24.1	23.9	21.7	18.3	14.9	8.5		
		MCDB	10.4	14.2	21.2	27.4	30.7	33.7	34.8	34.1	31.8	26.8	20.5	11.1		
	2%	WB	6.0	8.4	13.3	18.0	20.0	21.9	23.0	22.7	20.5	16.8	12.5	6.7		
		MCDB	8.5	12.3	19.9	25.6	29.5	32.7	33.9	33.2	30.5	24.7	17.0	9.4		
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	5%	WB	4.7	7.4	12.4	16.8	19.1	21.1	22.2	21.9	19.3	15.7	11.2	5.6		
		MCDB	6.7	10.4	18.2	23.6	28.1	31.8	33.1	32.6	28.9	22.6	15.3	7.8		
	10%	WB	3.7	6.2	11.5	15.8	18.3	20.4	21.5	21.2	18.2	14.7	10.0	4.6		
		MCDB	5.5	8.8	16.5	22.2	26.6	30.8	32.1	31.5	27.3	20.7	13.3	6.7		
	5% DB	MDBR	6.8	7.7	9.4	10.8	12.2	13.6	13.5	13.7	13.5	11.6	9.3	7.1		
		MCDBR	9.2	10.3	12.5	13.8	14.8	15.6	15.0	15.0	15.1	14.1	12.1	10.0		
(36) Mean Daily Temperature Range	5% WB	MCWBR	6.1	6.3	6.2	6.2	6.1	6.2	6.0	6.3	7.1	7.3	7.0	6.7		
		MCDBR	8.2	9.3	11.2	12.2	13.3	14.4	14.2	14.3	14.7	12.8	11.1	8.8		
	5% WB	MCWBR	5.7	5.8	5.9	5.9	6.0	6.1	6.1	6.5	7.2	6.8	6.5	6.1		
		MCDBR	8.2	9.3	11.2	12.2	13.3	14.4	14.2	14.3	14.7	12.8	11.1	8.8		
(40) Clear-Sky Solar Irradiance	taub		0.347	0.397	0.467	0.495	0.492	0.494	0.483	0.448	0.433	0.466	0.417	0.350		
	taud		2.222	2.085	1.937	1.915	1.977	1.982	2.020	2.113	2.089	1.938	2.030	2.203		
	Ebn at Noon		788	792	774	784	797	794	799	817	799	702	683	746		
	Edh at Noon		99	131	170	186	179	178	170	151	144	150	120	94		
(44) All-Sky Solar Radiation	RadAvg		2.00	2.84	4.04	5.39	6.48	7.40	7.48	6.82	5.59	3.68	2.35	1.77		
	RadStd		0.18	0.31	0.29	0.46	0.39	0.29	0.24	0.27	0.18	0.32	0.29	0.22		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	+0.36	N/A	N/A	+0.51	N/A	N/A	N/A	+115	+81	
(47)	Regional (7 neighbors)	N/A	N/A	N/A	+0.51	N/A	N/A	N/A	+75	+64	

Nomenclature: See separate page